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DATAWATCH

COVID-19 Vaccination Of People Experiencing Homelessness And Incarceration In Minnesota

We used data from a statewide public health–health system collaboration to describe trends in COVID-19 vaccination rates by racial and ethnic groups among people experiencing homelessness or incarceration in Minnesota. Vaccination completion rates among the general population and people incarcerated in state prisons were substantially higher than those among people experiencing homelessness or jail incarceration.

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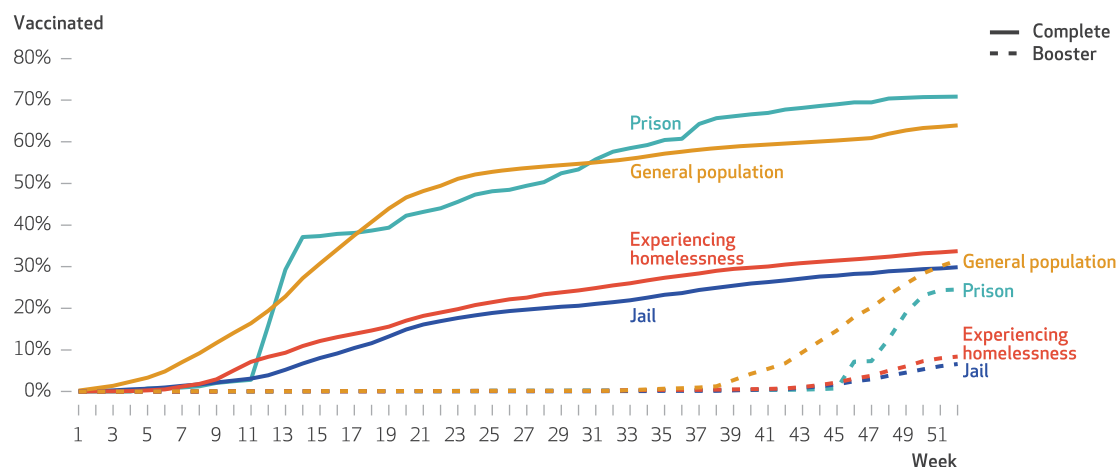
As COVID-19 has spread throughout the US, large outbreaks and high transmission rates have been documented in congregate settings including homeless shelters, jails, and prisons.^{1,2} Vaccines remain a key method to reduce the spread of COVID-19 and prevent serious illness, hospitalization, and death from COVID-19, but barriers limit vaccine access and acceptance among homeless and incarcerated populations.³

We characterized COVID-19 vaccination rates

among people in Minnesota who experienced homelessness or incarceration, using a novel statewide distributed data model developed by health systems and the Minnesota Department of Health. As of the end of 2021, vaccination completion rates among the general population (64.0 percent) and people incarcerated in state prisons (70.9 percent) were substantially higher than among people experiencing homelessness (33.7 percent) or jail incarceration (29.9 percent) (exhibit 1). Booster shots among people experiencing homelessness or incarceration also

EXHIBIT 1

Cumulative completed COVID-19 vaccination rates and booster vaccination rates among risk groups in Minnesota, by week, 2021



SOURCE Authors' analysis of data from the MN EHR Consortium. **NOTE** Weeks are MMWR weeks, a standardized epidemiological date system used by the Centers for Disease Control and Prevention.

lagged substantially compared with the general population. These data highlight the urgency of targeted outreach to populations in congregate settings amid the resurgence of SARS-CoV-2 variants.

Study Data And Methods

We matched data from the MN EHR Consortium, Minnesota Immunization Information Connection, Minnesota's Homeless Management Information System, and Minnesota Department of Corrections from 2021 (the data sources and the matching procedure are described in the appendix).⁴ We identified people who received services indicative of experiencing homelessness or housing instability (outreach, shelter bed use, subsidized hotel stays, and single-site transitional or permanent supportive housing); we refer to them as having experienced homelessness. We generated non-mutually exclusive indicators for people who experienced homelessness or jail or prison incarceration from September to November 2021. Complete vaccination was defined as receiving at least two doses of the BNT162b2 (Pfizer-BioNTech) or mRNA-1273 (Moderna) COVID-19 vaccines or one dose of the Ad26.COV.2.S (Janssen) vaccine. Boosters were defined as any vaccine received after the second Pfizer-BioNTech or Moderna vaccine or the first Janssen vaccine.

Patient age, race and ethnicity, sex, preferred language, and common comorbidities were obtained from electronic health record (EHR) data.

For the general population, we calculated vaccination rates by dividing the number of people who received a complete vaccination (one or two doses, depending on the vaccine) by Census Bureau population estimates. We then plotted trends in vaccination rate per *MMWR* week from January through December 2021. *MMWR* weeks are a standardized epidemiological date system used by the Centers for Disease Control and Prevention. Data aggregation and analysis were performed in R, version 4.1.1.

Because populations experiencing homelessness or jail incarceration can be highly transient, we focused on people who experienced homelessness or jail or prison incarceration between September and November to estimate vaccination rates in populations currently in congregate settings. Thus, the trends we observed might not capture early vaccination outreach efforts. Because the prison population is more stable than the jail or homeless cohorts, trends in this population captured targeted vaccination efforts with greater fidelity in early study periods.

Furthermore, our homelessness indicator may have underestimated the total homeless popula-

tion—particularly people who are unsheltered or not using services—because it captured only those who accessed homeless services in the Homeless Management Information System. Estimates among the entire population of people experiencing homelessness may differ if vaccination rates differ between people who do and do not use services. Last, data were limited to people with past health system interactions. Internal validity checks showed demographic similarities between matched and unmatched individuals; however, we could not rule out unobserved differences resulting in different vaccination rates among unmatched individuals.

Study Results

CUMULATIVE VACCINATION RATES By the end of 2021, 64.0 percent of the general population in Minnesota and 70.9 percent of people recently incarcerated in prison had completed the COVID-19 vaccine series (exhibit 2). This far exceeded the vaccination rate among people experiencing homelessness (33.7 percent) or jail incarceration (29.9 percent). Trends in vaccination over time showed differing patterns of vaccine uptake between the homeless, jail, prison, and general populations (exhibit 1). During the first half of 2021, vaccination rates among the general population steadily rose, reaching 52.8 percent by week 25 before slowing. Rates also rose steadily in the homeless and jail populations during this period, although vaccine uptake remained much lower. In contrast, prison vaccination rates increased sharply during spring 2021, when rates rose from 2.8 percent in week 11 to 37.4 percent in week 15.

The homeless and incarcerated populations did not experience an increase in boosters until November, eleven weeks after the general population showed an increase. By the end of 2021, the cumulative rate of boosters in the general population (31.5 percent) was substantially higher than in the homeless (8.4 percent), jail (6.6 percent), and prison (24.5 percent) populations (exhibit 1).

VACCINATION RATES

► **DEMOGRAPHIC CHARACTERISTICS:** People ages sixty-five and older had the highest cumulative rates of vaccination across all groups (exhibit 2). Rates were higher among females compared to males in the general (66.6 percent versus 60.2 percent) and jail (31.4 percent versus 29.3 percent) populations, but not the homeless population (28.4 percent versus 37.9 percent). Among homeless and incarcerated populations, people with comorbidities had vaccination rates higher than those of the overall homeless and incarcerated populations.

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EXHIBIT 2

Cumulative COVID-19 vaccination rates and demographic characteristics among Minnesotans, by risk group, weeks 1–48, 2021

	Cumulative vaccination rates, by population			
	Homelessness (n = 13,236)	Jail incarceration (n = 11,994)	Prison incarceration (n = 7,164)	General (n = 5,563,378)
Overall vaccination rate	33.7%	29.9%	70.9%	64.0%
Age, years				
0–11	4.0	— ^a	— ^a	16.6
12–18	23.6	— ^a	— ^a	52.6
19–24	22.3	20.7	75.5	55.0
25–44	31.1	26.5	70.8	67.6
45–64	50.0	41.7	69.5	73.8
65+	51.7	56.5	72.3	99.6
Race and ethnicity				
American Indian	36.9	35.8	77.8	46.9
Asian or Pacific Islander	34.1	46.5	74.1	70.3
Black	28.4	23.5	64.6	50.4
Hispanic	33.1	27.0	73.2	53.5
White	39.4	31.5	75.8	62.2
Multiple races	26.6	29.0	64.7	— ^b
Other or missing	25.8	23.3	60.7	— ^b
Sex				
Male	37.9	29.3	70.9	60.2
Female	28.4	31.4	71.0	66.6
Preferred language				
English	33.4	29.7	71.0	— ^b
Spanish	43.6	28.6	72.6	— ^b
Somali	30.4	— ^c	— ^c	— ^b
Other	43.4	40.6	71.7	— ^b
Missing	44.4	32.9	69.9	— ^b
Comorbidities				
Asthma	41.1	33.6	71.9	— ^b
Chronic obstructive pulmonary disease	46.9	35.1	71.7	— ^b
HIV	67.2	43.5	90.0	— ^b
Cancer	52.4	50.4	78.1	— ^b
Heart disease	57.9	47.7	72.5	— ^b
Diabetes	56.1	50.7	75.0	— ^b
Hypertension	55.3	44.1	79.3	— ^b
Chronic kidney disease	60.9	50.5	82.0	— ^b
Substance use disorder	45.7	32.5	75.7	— ^b

SOURCE Authors' analysis of data from the MN EHR Consortium. **NOTE** All categories of race and ethnicity except "Hispanic" are non-Hispanic. ^aAggregated data include information on 258 and 118 minors (age 18 or younger) in the jail and prison populations, respectively. We suppressed vaccination rates for minors experiencing incarceration as they might not be representative of all minors with involvement in the criminal justice system. ^bUnlike the homelessness, jail, and prison populations, denominator estimates for the general population are derived from Census Bureau population estimates, the American Community Survey 5-year data, and Minnesota Compass and not directly from patient records. For the general population we suppressed rates by multiple or other and unknown races, language, and comorbidities, as the measurement of the numerator and denominator are inconsistent. ^cSample sizes less than 11 were suppressed in accordance with privacy policies.

► **RACE AND ETHNICITY:** Non-Hispanic Asian people had the highest vaccination rates in the general (70.3 percent) and jail (46.5 percent) populations (exhibits 3 and 4). Non-Hispanic White people had the highest rate in the homeless population (39.4 percent), and non-Hispanic American Indian people had the highest vaccination rate in the prison population (77.8 percent) (exhibits 5 and 6). Across all populations, the vaccination rate for non-Hispanic White people (31.5–75.8 percent) exceeded the rate for non-Hispanic Black people (23.5–

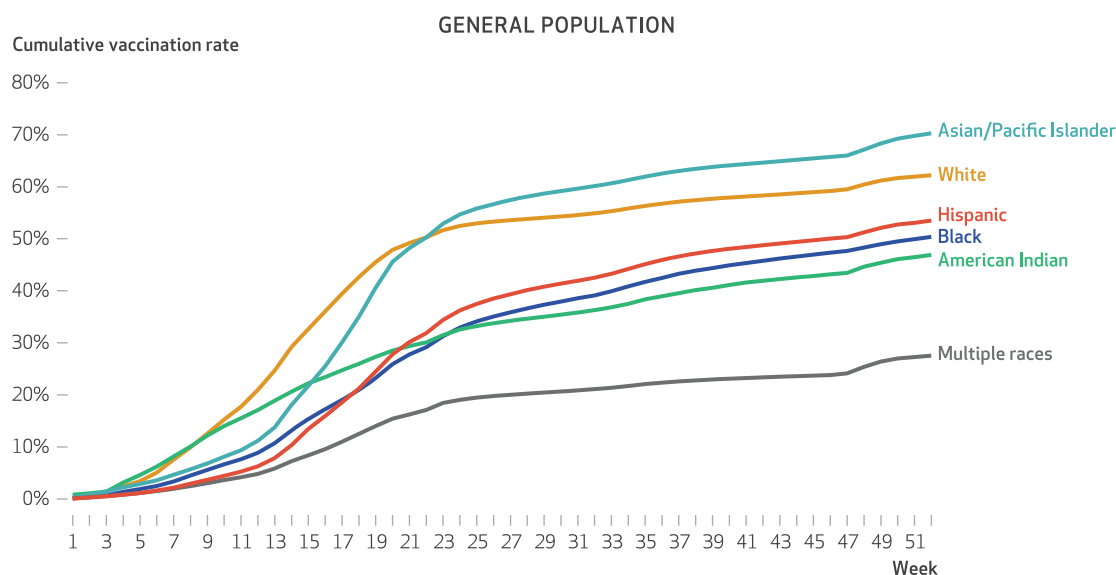
64.6 percent) and Hispanic people (27.0–73.2 percent).

Discussion

In this study we characterized COVID-19 vaccination rates for high-risk populations, using EHR data linked to statewide data on vaccination, homelessness, and incarceration. We found that the homeless and jail populations had significantly lower vaccination rates compared with the general population. Concerningly, these

EXHIBIT 3

Cumulative COVID-19 vaccination rates among racial and ethnic groups in the Minnesota general population, by week, 2021



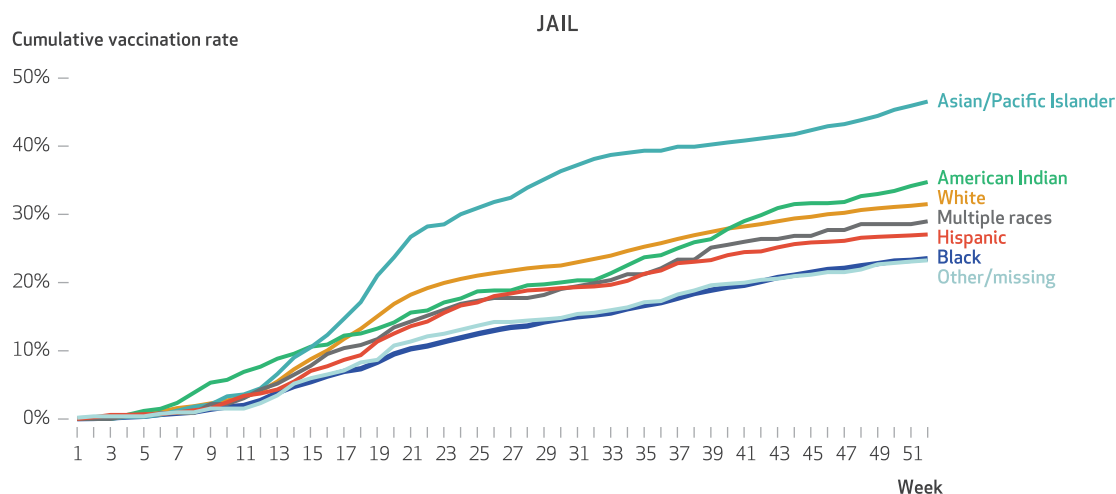
SOURCE Authors' analysis of data from the MN EHR Consortium. **NOTES** Weeks are MMWR weeks, as defined in the exhibit 1 notes. All categories of race and ethnicity except "Hispanic" are non-Hispanic.

inequities have persisted and may be worsening through vaccine booster administration. In Minnesota, the general population has received a booster at four times the rate of the homeless and jail populations. These findings are worrisome, as people living in congregate settings, compared with the general population, are at higher risk for COVID-19 transmission because

of challenges in practicing social distancing and taking other precautions.^{1,2} Furthermore, people transitioning between congregate living settings and the community have been linked to increased community transmission.⁵ This emphasizes the importance of increasing vaccination rates among people experiencing homelessness or incarceration as novel SARS-CoV-2 variants

EXHIBIT 4

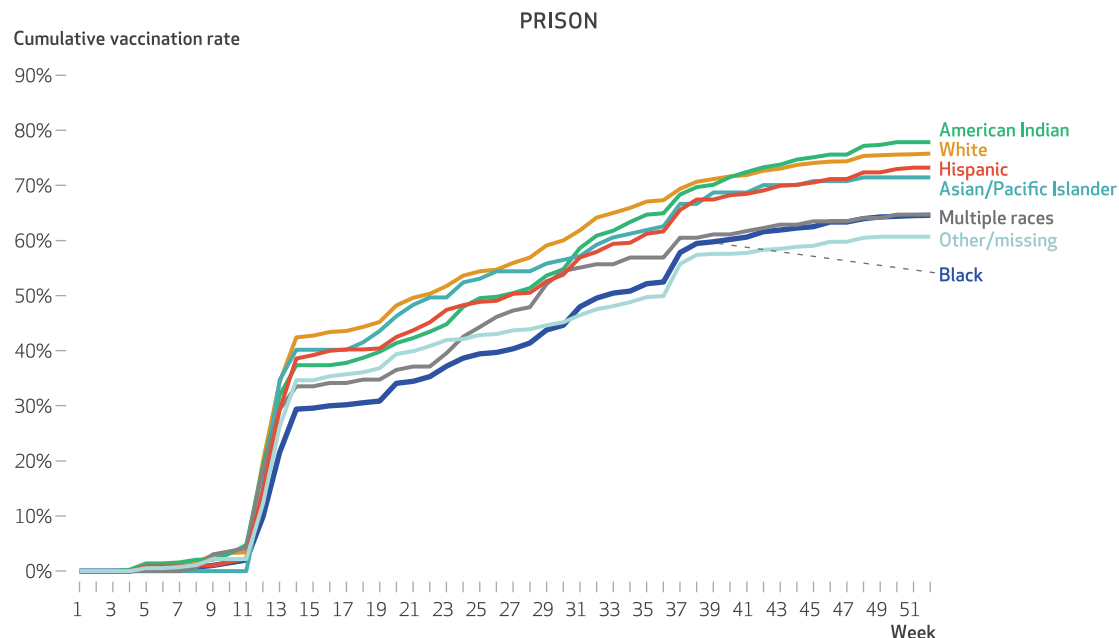
Cumulative COVID-19 vaccination rates among racial and ethnic groups in the Minnesota jail incarceration population, by week, 2021



SOURCE Authors' analysis of data from the MN EHR Consortium. **NOTES** Weeks are MMWR weeks, as defined in the exhibit 1 notes. All categories of race and ethnicity except "other/missing" and "Hispanic" are non-Hispanic.

EXHIBIT 5

Cumulative COVID-19 vaccination rates among racial and ethnic groups in the Minnesota prison incarceration population, by week, 2021



SOURCE Authors' analysis of data from the MN EHR Consortium. **NOTES** Weeks are MMWR weeks, as defined in the exhibit 1 notes. All categories of race and ethnicity except "other/missing" and "Hispanic" are non-Hispanic.

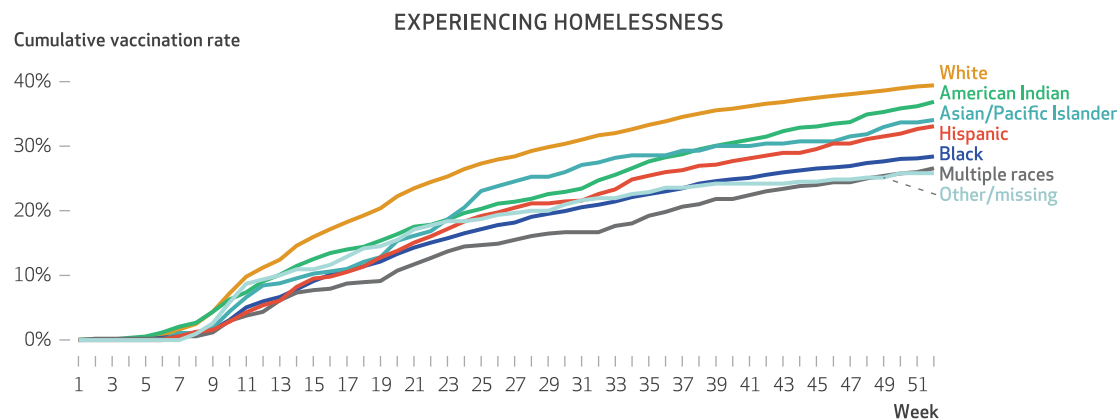
continue to emerge.

In this study we identified substantially different trends in COVID-19 vaccination between prison and jail populations despite similar exposure risks. By the end of 2021, cumulative vaccination rates for Minnesota's prisons (70.9 percent) were higher than previous national

findings from the Federal Bureau of Prisons (64.2 percent of people offered a vaccine in a Federal Bureau of Prisons institution accepted it).⁶ To our knowledge, no existing research has documented vaccination rates among jail populations specifically, but the low rates (29.9 percent) we describe in this study indicate an urgent

EXHIBIT 6

Cumulative COVID-19 vaccination rates among racial and ethnic groups in the Minnesota homelessness population, by week, 2021



SOURCE Authors' analysis of data from the MN EHR Consortium. **NOTES** Weeks are MMWR weeks, as defined in the exhibit 1 notes. All categories of race and ethnicity except "other/missing" and "Hispanic" are non-Hispanic.

We found that the homeless and jail populations had significantly lower vaccination rates compared with the general population.

public health need. Given low vaccination rates in jails and the impact they have on community transmission rates, additional resources and new processes will be needed to overcome operational challenges in jail settings that impede vaccination efforts, such as brief stays and medical staff shortages.^{3,5} For example, to accommodate shorter stays, COVID-19 vaccinations could be offered to people upon their entry into jail.

We found the vaccination rate among Minnesota's homeless population (33.7 percent) to be substantially lower than among the general population (64.0 percent). This stark inequity is consistent with findings from six cities indicating that vaccination rates among people experiencing homelessness range between 22 percent and 52 percent.⁷

Among all populations, non-Hispanic Black people and Hispanic people received COVID-19 vaccines at lower rates than non-Hispanic White people. This finding is consistent with previous research indicating disparities in vaccination rates between Black and White people in prison settings and the general population.^{6,8} Structural racism and historic abuses in medicine, research, and the criminal justice and housing sys-

tems have likely contributed to racially disparate vaccination hesitancy and uptake.^{9,10} However, rates of vaccine acceptance in a prison setting improved substantively when people were offered the vaccine for a second time.⁶ Continued data to monitor vaccination rates are essential to ensure that investments in vaccine outreach lead to uptake among high-risk populations.

As new SARS-CoV-2 variants emerge, boosters may play an increasingly important role in reducing infection and severe disease.¹¹ However, we found persistent inequities between the general population and those experiencing homelessness or incarceration and substantial racial and ethnic disparities within populations. Departments of health and community organizations emphasized policies and initiatives to mitigate vaccination inequities early in the pandemic.¹² More than a year since the first American received a COVID-19 vaccine, these organizations must continue efforts such as outreach and prioritized access to reduce inequities in booster vaccination.

Conclusion

Accurate and timely data can support the prioritization of high-risk populations during an evolving crisis, but reliable data for populations experiencing homelessness or incarceration are often difficult to obtain. During the COVID-19 pandemic, heterogeneity in state-level vaccine allocation strategies has reinforced this longstanding need for timely and hyperlocal data to inform the equity-driven prioritization of high-risk groups.^{1,2} In Minnesota we integrated data for people experiencing homelessness or incarceration with statewide vaccination and EHR data to delineate trends in vaccination rates. As new SARS-CoV-2 variants emerge, the use of real-time data can benefit all states by informing continuous outreach to mitigate inequities in COVID-19 vaccination rates. ■

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